

Fundus Image Analysis for Age related Retinal Disease Detection using Deep Learning and XAI Methods

Umadevi Venkatappa

*Department of Computer Science and Engineering
B.M.S. College of Engineering
Bengaluru, India
umadevi.cse@bmsce.ac.in*

Somnath Mukherjee

*Pixelish System
Bengaluru, India
sommnath@pixelish.com*

Shrinanda Shivprasad Dinde

*Department of Computer Science and Engineering
B.M.S. College of Engineering
Bengaluru, India
shrinanda.cs23@bmsce.ac.in*

Soham Hathi

*Department of Computer Science and Engineering
B.M.S. College of Engineering
Bengaluru, India
sohamhathi.cs23@bmsce.ac.in*

Sinchana

*Department of Computer Science and Engineering
B.M.S. College of Engineering
Bengaluru, India
sinchana.cs23@bmsce.ac.in*

Samridhhi Singh

*Department of Computer Science and Engineering
B.M.S. College of Engineering
Bengaluru, India
samridhisingh.cs23@bmsce.ac.in*

Abstract— Early recognition of age-related retinal conditions is crucial in avoiding irreversible loss of vision. This study presents a Deep Learning based diagnostic pipeline for classification of retinal fundus images. The RFMiD 1.0 and 2.0 datasets were merged and standardized, to produce a combined dataset consisting of 1098 fundus images covering ten different classes of age-related retinal diseases. All multilabel samples were discarded to keep a clean multiclass setting, and images were preprocessed via deterministic resizing, normalization, contrast enhancement and illumination correction. Trained four deep learning models ResNet-34, DenseNet-121, VGG-19 and Inception-V4. The models were evaluated using accuracy, precision, recall, F1-score and ROC-AUC score. Additionally, explainable AI techniques such as LIME, SHAP, Grad-CAM, and RISE were applied to interpret model predictions by highlighting pathology-relevant retinal regions. Among all the architectures, DenseNet-121 achieved the highest test accuracy of 88%, demonstrating the effectiveness of the proposed pipeline for reliable, interpretable, and scalable retinal disease screening.

Keywords— *Age-related Eye Diseases, CNN, Deep Learning, Explainable AI, Medical image Classification, Retinal Fundus Imaging*

I. INTRODUCTION

Age-related retinal diseases including age-related macular degeneration (ARMD), diabetic retinopathy (DR), branch retinal vein occlusion (BRVO) and central Retinal Vein occlusion (CRVO) are major causes of vision loss throughout the world. These conditions progress gradually showing few or no symptoms in their early stages, making timely detection necessary for preventing long-term visual impairment. World Health Organization reports, more than 2.2 billion people globally experience near or distance-vision impairment [1], and India alone contributes over 70 million people with such conditions [2]. With the rapid progress in deep learning [3] and medical imaging, convolutional neural networks (CNNs)

have been very successful in classifying complex retinal abnormalities from fundus images [4]. Retinal fundus imaging remains the most cost-effective approach for visualizing essential retinal structures including the macula, optic disc, and blood vessels for fast screening, especially in low-resource situations [5].

In this study, a single standardized experimental pipeline was designed to ensure a fair comparison between models. Four widely used CNN architectures—ResNet-34 [6], DenseNet-121 [7], VGG-19 [8], and Inception-V4 [9] were trained on the combined RFMiD dataset under identical experimental settings.

II. RELATED WORKS

Extensive research has been conducted on deep learning-based methods for the automatic detection of retinal fundus diseases. The work presented in [10] highlighted the fragmented nature of existing retinal disease research. This review examines various deep learning techniques used recently to diagnose five main eye conditions: diabetic retinopathy, cataract, age-related macular degeneration, glaucoma, and retinopathy of prematurity.

To address dataset inconsistencies, a self-supervised learning approach on the Age-Related Eye Disease Studies (AREDS) dataset for age-related macular degeneration (AMD) was proposed in [11]. Their method reduced dependency on manual annotations. An unbalanced accuracy of 94% was achieved using self-supervised learned features, while an accuracy of 95.8% was achieved by the supervised trained CNN.

To deal with the computational cost of Deep Learning (DL) models, a lightweight CNN with ten convolutional layers was presented in [12]. The model achieved 95% validation accuracy

on the EyeNet dataset. This demonstrates the feasibility of retinal diagnostics on resource-limited hardware.

According to the review in [13], picture quality, eye position, laterality, phase, and five types of DR-related lesions were produced by the Artificial Intelligence (AI) system with an accuracy ranging from 80.79% to 93.34%. Additionally, it was reported the DeepSeeNet algorithm outperformed retinal specialists in patient-based classification (accuracy=0.671, kappa=0.558; accuracy=0.599, kappa=0.467, respectively), with a high AUC in detecting large drusen (0.94), pigment abnormalities (0.93), and late AMD (0.97). In [14], a lightweight augmented CNN integrated with Gradient-Weighted Class Activation Mapping (Grad-CAM) and Grad-CAM++ was presented. The Adam optimizer with learning rate of 1×10^{-4} was used. The 16.6M parameter network converged in fewer than 50 epochs and reached 87.9% test accuracy, a macro precision of 88.2%, a macro-recall of 87.9%, and a macro-F1-score of 88%, reducing the error by 58%.

A hybrid Generative Adversarial Network (GAN)/CNN transfer learning pipeline for predicting AMD progression was proposed in [15]. Their results showed the use of multi-stage deep learning models in clinical decision support. The Study [16] developed a model on dataset containing 4303 retinal images, representing four classes of pathologies. The proposed embedded model achieved an accuracy of 98.06% on two combined datasets collected from Ocular Disease Intelligent Recognition (ODIR) and Retinal Fundus Multi-Disease Image Dataset (RFMiD). In [17], a thorough review of AI applications in different retinal illnesses was presented, emphasizing improvements in patient outcomes, increased screening efficiency, and early detection.

Altogether, these works demonstrate that the DL models are effective for the classification of retinal fundus images. Despite these tremendous successes, the issues related to imbalanced datasets and model interpretability still exist. This limitation motivates the present study, which focuses on data augmentation and consistent evaluation across multiple CNN architectures to improve fairness and robustness in multi-class retinal disease classification.

III. METHODOLOGY

The process pipeline for classification of retinal fundus images is shown in Fig.1. The components of the pipeline are discussed in the subsequent subsections.

A. Dataset Merging

Two publicly available retinal fundus image databases, RFMiD1.0 [18] and RFMiD2.0 [19], were combined, which contains high resolution images captured under clinical settings and included expert-verified diagnostic labels. To maintain consistency across the dataset, all metadata fields and disease annotations were standardized, and naming conventions were unified.

B. Dataset Cleaning and Split

Since the objective of the study was to perform single-label multi-class classification, images containing multiple co-existing retinal conditions were dropped. After dataset cleaning, the final dataset consisted of 1,098 images belonging

to ten age-related retinal disease classes. The ten class labels are: Anterior Ischemic Optic Neuropathy (AION), ARMD, BRVO, CRVO, Drusen (DN), DR, Epiretinal Membrane (ERM), MHL (Macular Hole), Optic Disc Cupping (ODC), and Retinal Pigment Epithelium Changes (RPEC). The dataset was divided in a 70:15:15 ratio, resulting in 768 training images, 165 validation images, and 165 test images. The number of images in the training, validation, and test sets is shown in Table I.

C. Preprocessing Pipeline

All the images are resized to a uniform size to fit the model input. Normalization and illumination correction were applied to scale pixel values using ImageNet statistics and to reduce lighting variation in retinal images, respectively.

D. Training Pipeline

Baseline training (without augmentation): The model was trained on the raw dataset without doing any image transformations. The main purpose for doing this was to establish a baseline performance for comparison.

Hybrid augmentation (Fixed+Online): Data augmentation [20] increased dataset diversity by transforming the images. Fixed augmentation strategy was combined with online augmentation so that the minority disease classes are not unrepresented.

TABLE I. PER CLASS DISTRIBUTION OF FUNDUS IMAGES

Class	Train Count	Val Count	Test Count	Augmented Train Count
AION	15	4	3	45
ARMD	52	12	11	104
BRVO	69	15	15	69
CRVO	35	7	8	70
DN	111	23	24	111
DR	329	71	71	329
ERM	11	2	2	33
MHL	8	2	2	24
ODC	120	25	26	120
RPEC	18	4	3	54

TABLE II. MODEL PARAMETERS BEFORE AUGMENTATION

Model	Input Size	Architecture Type	Parameters
ResNet34	224×224	Residual (skip connections)	~21.8M
VGG19	224×224	Deep sequential conv layers	143.7M
DenseNet121	224×224	Dense connections (feature reuse)	~7.9M
Inception-v4	299×299	Multi-branch Inception modules	~42M

E. Deep Learning Models Training

Four deep learning networks (ResNet34, VGG19, DenseNet121 and Inception-v4) were finetuned for the classification of retinal fundus images in identical training

settings. All the models were first trained for 50 epochs with a fixed learning rate of 1×10^{-4} , batch size of 32 using the AdamW optimizer and CrossEntropyLoss.

All of the model's experiments were conducted with identical pre-processing and data set splits. These architectures are mainly different in terms of depth, connectivity pattern, input size constraints and the cost of computation.

To improve the class balance in the training set, particularly for minority disease classes such as AION, ERM, MHL, and RPEC fixed rotation-based augmentation ($\pm 9^\circ$) was applied. This procedure expanded AION, ERM, MHL, and RPEC by 200%, while moderately represented categories such as ARMD and CRVO were increased by 100%. The remaining well-represented classes such as BRVO, DN, DR, and ODC—were left unaugmented to avoid unnecessary redundancy. After fixed augmentation, the training set increased from 768 to 959 images. Table I shows number of images in final augmented training set.

TABLE III. MODEL PARAMETERS AFTER AUGMENTATION

Parameter	ResNet34	VGG19	DenseNet121	Inception-v4
Input Size	384×384	224×224	384×384	299×299
Architecture Type	Residual	Deep Fully Connected (FC) Layer + Conv Stack	Dense Connections	Multi-branch Inception
Training Strategy	Full fine-tuning	2-Stage (head → full)	Partial fine-tuning (last block)	Full fine-tuning
Regularization Used	Dropout(0.4)	VGG classifier + Dropout	BN + Dropout(0.5)	Dropout(0.4)
Classifier Head	Dropout → Linear	Large FC Stack	Batch Normalization (BN) → Dropout → Linear	Dropout → Linear
Parameters	~21M	143M	~8M	~42M

Online augmentations include random horizontal flipping, $\pm 20^\circ$ rotations, and color jittering were applied dynamically to introduce continuous variation in image appearance and prevent overfitting. During training, online augmentation was applied for training and validation images. This hybrid (Fixed+Online) augmentation strategy ensured that the models were exposed to balanced class distribution and diverse transformations on images. Model parameters used for training before augmentation and *after* augmentation are listed in Table II and III.

F. Evaluation Matrix

Certain metrics were used to check how well the model performs: Accuracy that gives percentage of correctly classified images, Precision that tells how reliable positive prediction are there, Recall that tells how many disease classes are correctly identified and F1-score that gives the balanced measure of precision and recall. Macro-Average F1 score to evaluate models' performance across all 10 classes. AUC-ROC (Area Under Curve-Receiver Operating Characteristic) score to evaluate the sensitivity of the classification model.

G. Explainability

XAI methods mainly identify the regions in fundus images that influenced the prediction. Each of the four models was analyzed using four different XAI techniques.

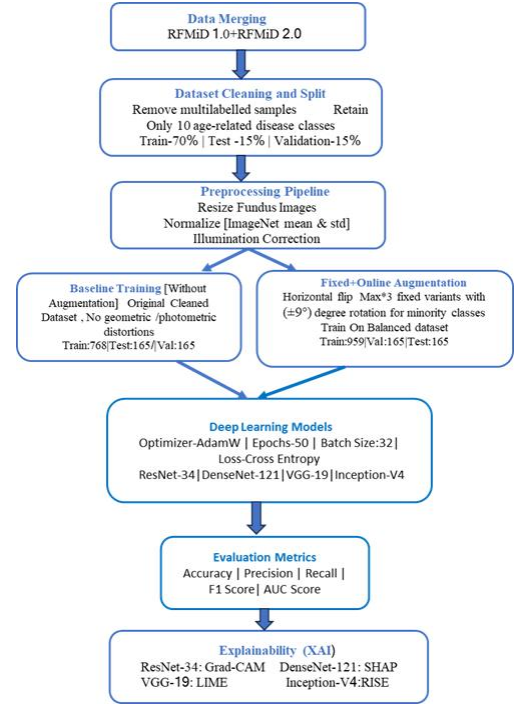


Fig. 1. Process pipeline for classification of retinal fundus images

IV. RESULTS AND DISCUSSION

In-depth analysis of four deep learning architectures—ResNet34, DenseNet121, Inception-v4, and VGG19—before and after applying the augmentation was carried out.

TABLE IV. OVERALL PERFORMANCE COMPARISON OF FOUR MODELS WITHOUT AUGMENTATION

Model	Accuracy (%)	Weighted Precision (%)	Weighted Recall (%)	Weighted F1-Score (%)	Macro F1-Score
ResNet-34	70.91	66.68	70.91	67.97	0.3848
DenseNet-121	72.73	71.75	72.73	71.58	0.5123
VGG-19	77.58	75.48	77.58	76.14	0.4608
Inception-v4	81.82	80.81	81.82	80.96	0.5657

The analysis highlights the influence of dataset imbalance, the effect of augmentation, and the final comparative performance of all the models.

Table IV summarizes the baseline results obtained from original imbalanced dataset before applying augmentation. Inception-v4 performed the best among all the models with test accuracy of 81.82%. It showed strong performance on high-frequency classes including DR, ODC and ARMD with F1-scores above 0.80. Due to imbalance in the RFMiD dataset, all models failed on rare disease classes like ERM and RPEC. Validation curves confirmed early and strong overfitting particularly in VGG19 and ResNet34, where the validation

accuracy ceases to increase and fluctuates across the epochs. Overall, the pre-augmentation performance revealed the inability of all models to generalize well across minority disease categories and models were biased towards majority classes.

TABLE V. OVERALL PERFORMANCE OF FOUR MODELS WITH AUGMENTATION

Model	Accuracy (%)	Weighted Precision (%)	Weighted Recall (%)	Weighted F1 (%)	Macro F1
ResNet-34	85.45	83.52	85.45	84.14	0.6480
DenseNet-121	88.48	88.74	88.48	87.32	0.6974
VGG-19	76.97	79.83	76.97	77.34	0.5991
Inception-v4	83.03	82.26	83.03	81.97	0.5957

After applying the fixed rotation-based ($\pm 9^\circ$) augmentation combined with online augmentation (flips, jitter, $\pm 20^\circ$ rotation), all models achieved substantial improvements in class-wise fairness, stability, and generalization. Table V reports an improvement in performance in ResNet34, with test accuracy rising from 70.91% $\rightarrow$ 85.45% and macro-F1 increasing from 0.38 $\rightarrow$ 0.6480. DenseNet121 achieved the best overall performance, improving test accuracy from 72.73% $\rightarrow$ 88.48% with macro-F1 rising from 0.5123 $\rightarrow$ 0.6974, and as shown in the Fig. 2 and 3 previously undetectable classes like ERM and RPEC became learnable. Inception-v4, which already performed well before augmentation, showed a modest accuracy gain of +1.21% and exhibited marked improvements in stability. VGG19 benefited substantially from reduced overfitting and smoother convergence, reaching 77.00% accuracy and increasing macro-F1 from 0.4608 $\rightarrow$ 0.599.

TABLE VI. PER CLASS F1-SCORE OF DENSENET-121

Classes	F1-Score (Before Augmentation)	F1-Score (After Augmentation)
AION	0.5000	0.5000
ARMD	0.4762	0.8421
BRVO	0.5926	0.8462
CRVO	0.8750	0.9412
DR	0.8652	0.9396
MHL	0.5000	0.0000
ERM	0.0000	0.6667
ODC	0.7541	0.9231
RPEC	0.0000	0.5000
DN	0.5600	0.8148

The results confirm that, addressing class imbalance is essential for developing clinically reliable deep learning systems. Fig. 4 shows that AUC-ROC curve for DenseNet-121 model which achieved a micro-Average AUC score of 97.7%. Table V reports an improvement in performance in ResNet-34. The analysis of the curves reveals that the model is able to learn ten disease classes data appropriately. F1 score for each class before and after augmentation is provided in Table VI. Classes ERM and RPEC achieved increased F1 score of 0.6667 and 0.500 after augmentation. But for MHL class there was a decrease in F1 score after augmentation.

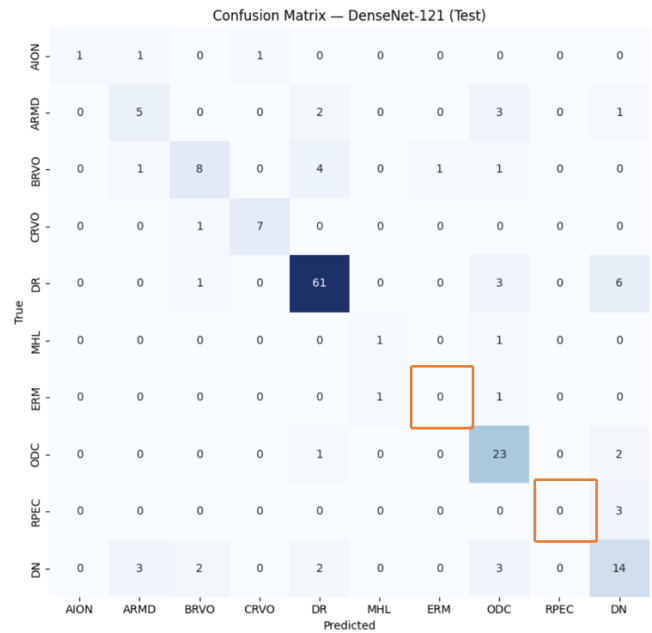


Fig. 2. Confusion Matrix for Dense-121 Before Augmentation

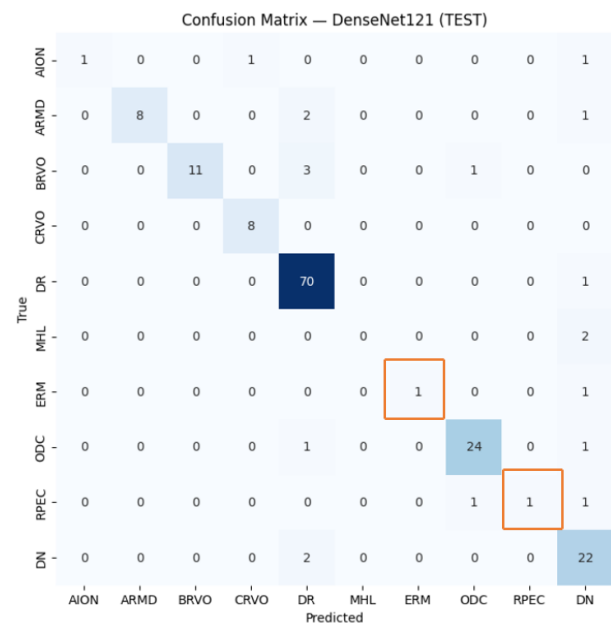


Fig. 3. Confusion Matrix for DenseNet-121 After Augmentation

1) Explainable AI (XAI) Integration

For improving interpretability XAI methods were used, including LIME for VGG-19, SHAP for DenseNet-121, RISE for Inception-V4, and Grad-CAM for ResNet-34. The visual explanation of all four techniques highlighted the retinal areas that majorly contributed toward the predictions of each model. Table VII shows XAI methods interpretation for four models for a sample image of DR class.

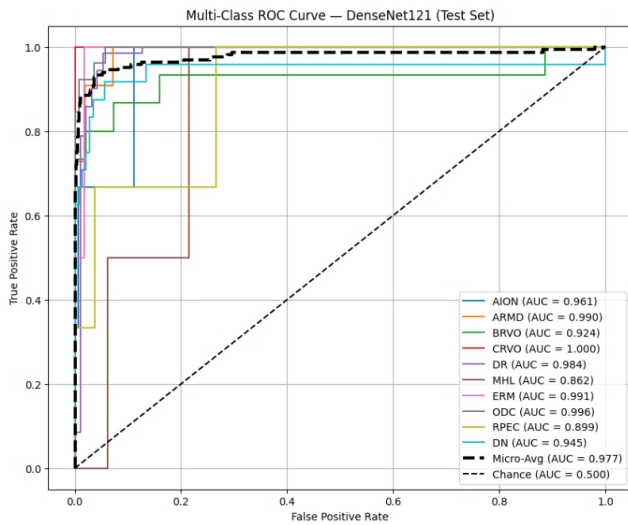


Fig. 4. ROC-AUC Curve for DenseNet-121

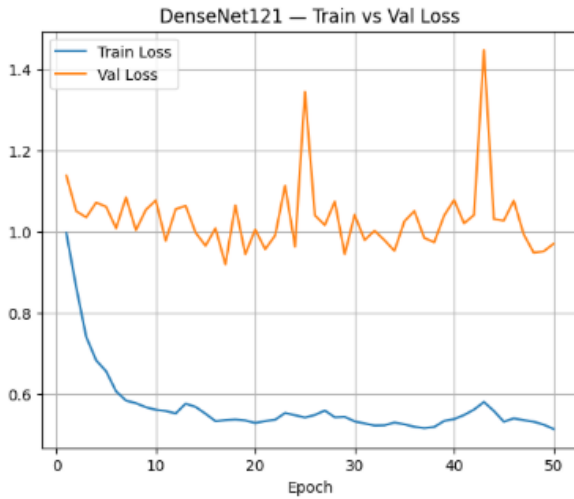


Fig. 5. Illustrates the Training and Validation Loss Curve for DenseNet121

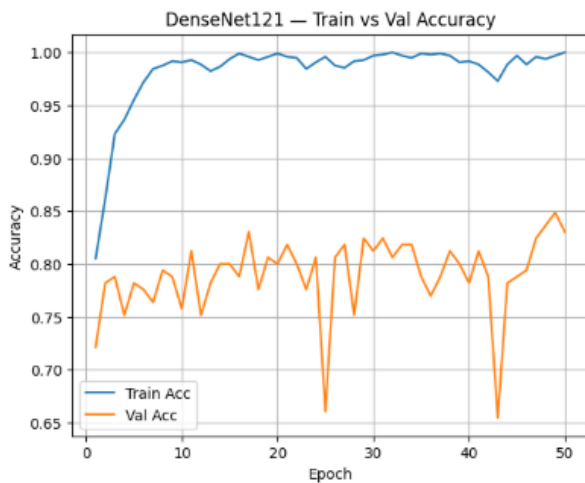


Fig. 6. Illustrates the Training and Validation Accuracy Curve for DenseNet121

TABLE VII. SAMPLE XAI IMAGES FOR FOUR MODELS

Model	XAI	Original Image	XAI Image
Resnet34	Grad-CAM		
InceptionV4	RISE XAI		
VGG19	LIME		
Densenet121	SHAP XAI		

V. CONCLUSION AND FUTURE WORK

This work explored Deep Learning application for detection of ten age-related retinal diseases using fundus images. By applying hybrid augmentation (Fixed+Online). Four CNN models, ResNet-34, DenseNet-121, VGG-19, and Inception-V4 were trained under identical experimental settings.

The results highlight the importance of managing dataset imbalance. Without augmentation, models could not predict rare diseases. After applying the hybrid augmentation strategy, every model showed improvements in generalization and minority-class detection. DenseNet-121 achieved the best results with 88% test accuracy and ResNet-34 85.45%, Inception-V4 83.03% and VGG-19 benefited from reduced overfitting and achieved 77% test accuracy.

To demonstrate the areas of the retina that the model focussed on, XAI techniques like LIME, SHAP, RISE, and Grad-CAM were implemented. XAIs can bridge the gap existing between algorithmic decisions and interpretability, so that the system becomes more transparent and acceptable for medical practice.

The classification result and explainability of all four models can be provided to a Large Language Models (LLM) to generate diagnosis report, which can be looked forward as a future work.

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